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Micromechanical modelling of cellulose hydrogel composites based on coarse-grained molecular dynamics

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The mechanical response and structure of cellulose hydrogel composites (CHCs) are modelled as cellulose microfibrils using a bead-spring coarse-graining approach. Our results demonstrate that varying the adhesive contact energy between the flexible chains, as well as the number of contacts and structural anisotropy, significantly impacts the mechanical response of the network structure under tensile forces. Specifically, decreasing the contact energy and increasing its range is sufficient to increase the network's extensibility while decreasing its overall modulus. This key finding aligns qualitatively with experiments where the inclusion of polysaccharides in bacterial cellulose-based CHCs had an analogous effect. We hypothesise that polysaccharides, including hemicelluloses, facilitate alignment under strain by increasing the range ("softening") of the contact forces between microfibrils. Conversely, their absence results in "hard" contacts between microfibrils that are more energetic and can only act over short distances. In contrast to finite element models for CHCs, this coarse-grained approach incorporates non-permanent contacts between flexible microfibrils that permits structural rearrangement of the network in response to deformation, whereby controlling the density and proportion of long- and short- range contact forces suffice to qualitatively describe experiments. Controlling interactions between microfibrils thus provides a lever for designing CHCs with specific mechanical properties for various applications. Additionally, we suggest that plants naturally tune these variables in plant cell development to balance wall rigidity and extensibility.

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1. Introduction

There is a growing interest in the study of cellulose-based hydrogels derived from plant cell wall polysaccharides, given their role as a key component of materials for food, paper,

biomaterials and, more recently, biofuel industries.¹ Similarly, cellulose hydrogels of bacterial origin are finding an increasing market in biomedical applications and high-performance nanomaterials.² When produced in the presence of hemicelluloses to tune their mechanical properties, the resulting bacterial cellulose hydrogel composites (CHCs) can display a complex micromechanical behaviour that is difficult to describe through fully atomistic models.³ Explaining such behaviour may also shed light on the interplay between the primary plant cell wall constituents, which continues to be an active area of inquiry in plant biology and material sciences.⁴ The earliest description of CHCs corresponds to the so-called "tethered network model" of the primary plant cell wall, in which a relatively well-aligned network of cellulose microfibrils (MFs) are tethered by hemicelluloses (most significantly xyloglucan (XG) in "type I" walls of dicots, gymnosperms and non-commelinid monocots) and embedded in a gel matrix which, for dicotyledonous plants, is primarily composed of pectin.^{5,6} The tethered network model of CHCs in the plant wall has numerically been implemented through both deterministic⁷ and stochastic^{8–10} approaches. Stochastically, the main approach consists of using finite

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element simulations to simulate the behaviour of large numbers of tethered MFs. Interestingly, the model by Puri *et al.*⁹ postulated that hydrogen bonding between MFs and XG must become a major load-bearing constituent to retain cell wall integrity even at the smallest strains ($\sim 1\%$) is contrary to thermodynamic estimates of the XG–cellulose bonding energy.¹¹ Since these small strains are routinely experienced due to variations in turgor pressure, it was concluded that another load-bearing mechanism¹² must be in place to maintain wall integrity.

More recently, Cosgrove and colleagues proposed the “hot-spots” model of the cell wall, which poses that pectin–MF interactions are prevalent, while XG mediates the contact between a limited number of MF sites that are specifically targeted by expansins to disrupt hydrogen bonding and thereby allow wall loosening.^{13,14} Cosgrove’s model arose as a response to the landmark work by Cavalier *et al.*,¹² who demonstrated that genetic deletion of the side chains of XG, thereby resulting in a “naked glucan chain (*i.e.* cellulose), had minor impact on growth and development in the model plant *Arabidopsis thaliana*, while leading to greater cellulose MF alignment and rearrangement of cell wall components to compensate for the absence of XG. Thus, it is implicitly considered that non-cellulosic polysaccharides act as modulators of cellulose–cellulose interactions. Stochastic simulations¹⁵ have been used to test the hotspots hypothesis, finding that the proposed fraction of biological hotspots (0.3% cell wall mass) leads to a wall stiffness far below that required to oppose turgor pressure. Since this model was based on point-like joints (analogous to the tethered network model), it is clear the “hot-spots” cannot be point-like but are spread over a small but significant MF surface. Recently, Pieczywek *et al.*¹⁶ summarized the key intrinsic limitations of finite element models of primary plant cell walls, which include treating cellulose fibres as rigid rods and representing inter-fibre bonds as permanent crosslinks without incorporating the critical processes of bonding–debonding, fibre sliding, and self-healing. This oversimplification restricts their ability to accurately simulate large deformations, stress relaxation, creep behaviour, and the nuanced contributions of other wall components like pectin. Therefore, it remains necessary to develop an alternative modelling framework that explicitly incorporates intimate contact between MFs spread over a significant contact length. We highlight, however, that finite element simulations continue to be extremely valuable to model cell and tissue-level phenomena.¹⁷ For instance, this method has been widely used to develop or fine-tune constitutive models of plant tissues from micromechanical experiments (*e.g.* nanoindentation¹⁸ or lab-on-a-chip bending measurements¹⁹) and, once an appropriate model is obtained, to simulate the evolution of cell/tissue volume and shape during growth (*e.g.* in pollen tubes,²⁰ root hairs²¹ or trichome epidermal cells²²).

For CHC networks based on bacterial cellulose (BC), nanocrystalline cellulose filaments aggregate and twist into ribbon-like fibres.^{23,24} Therefore, it is these ribbons that constitute the minimum indissoluble element within the network, as opposed to individual MFs. To the best of our knowledge, the hot-spots

model does not necessarily operate in BC hydrogel composites. However, *Gluconacetobacter xylinus* (formerly *Acetobacter xylinum*) based CHC hydrogels prepared in the presence of finely tuned amounts of hemicelluloses (XG, arabinoxylan (AX), glucan or glucomannan) and pectin^{25–28} provides an experimental means by which to quantify the influence of hemicelluloses on mechanical behaviour of cellulose hydrogels, thus providing data to test and validate against models with variations in interactive forces between MF constructs. This is anticipated to provide a basis for developing rational hypotheses for describing the structural design of the cell wall without the impact of biological features such as the middle lamella.⁴

We propose here a modelling approach using coarse-grained (CG) molecular dynamics (MD) methods. We incorporate short- and long-range attractive forces between semi-flexible chains (bead-spring) in a network structure, allowing contact over a finite MF length. Using experimentally informed parameters, this simple model aims to qualitatively replicate the behaviour of bacterial CHC under tension. Our approach differs from previous approaches in the following aspects:

(i) Individual fibres (either MFs or cellulose ribbons) can aggregate into thicker macrofibrils: ribbons constitute the dominant structural unit in bacterial CHCs and can continue to aggregate to reach fibre thicknesses in the order of ~ 100 nm.^{4,27,29} In the primary cell wall, at least some fraction of MF are known to form tightly bound bundles in the primary wall.^{30,31}

(ii) The resulting ribbons interact over finite length scales rather than as point contacts: there exists a finite contact length between fibres (for both MFs and ribbons). This representation is more realistic than discrete junctions employed in finite element models.^{8,10} Even if the attractive energy between beads is uniform, differences in contact length will generate a distribution of contact energies throughout the network.

(iii) The response of the structure to large strains can be studied: this is a computational limitation to finite element models of polymer networks since cellulose MFs interact through XG chains, past a certain strain the junctions must start to break, generating discontinuities.³⁰ This limitation has recently been addressed through the phase field formalism,³¹ but has not yet been applied to CHC systems.

Importantly, we have not explicitly incorporated XG in our simulations for two main reasons: (i) incorporating XG requires additional interaction parameters that are difficult to reasonably tune; and (ii) computational efficiency: physics-centred coarse-grained models based on the Martini force field can only efficiently tackle a small number of MFs (~ 20 – 40) of relatively short length (~ 50 nm) (see López *et al.*,³² Grünwald *et al.*³³). Bead-spring models of cellulose nanocrystal thin films, such as those proposed by Qin *et al.*³⁴ and Mehandzhysk *et al.*³⁵ can model hundreds of chains comprising hundreds of beads each (up to ~ 1 μ m long), but incorporating a secondary polymer with a much smaller weight per repeating unit would require adding many chains even at small wt%. Moreover, the large differences in chain diameter between the cellulose MFs and individual XG chains would require a considerable methodological change (for instance, considering that each bead on the

MF is comprised by several smaller particles). In addition, we do not explicitly consider water molecules, but use the implicit solvent formulation in the Langevin thermostat.³⁶ This means that its effects, such as viscous friction and Brownian collisions, are accounted for through a velocity damping and a random force term set to produce, on average, the desired target temperature.

Recently, Pieczywek framework to that in the recent bead-spring CG model proposed by Zhang *et al.*³⁷ who explicitly considers three types of beads (cellulose, XG and pectin) to precisely match the structure revealed in SEM micrographs of onion cell wall epidermis. Geitmann *et al.*³⁸ assessed the scope of this framework, underscoring how realistic molecular-level simulations can reconcile the traditionally puzzling combination of extensibility and high tensile strength in primary walls. Geitmann *et al.* analysis highlighted the importance of accurately portraying cellulose MF reorientation and the limited, though nontrivial, role of hemicelluloses in fine-tuning the wall's mechanical properties by mediating the contact between MFs. We take this conclusion a step forward by fully eliminating non-cellulosic polysaccharides, and to investigate the hypothesis that cellulose fibre-fibre contacts and orientation is sufficient to produce a qualitative description of the behaviour of bacterial CHC by implicitly considering XG in a swollen system.

2. Theoretical methods

2.1 Description of the model

Cellulose fibres (MFs or ribbons) are modelled as semi-flexible chains using a modified bead-spring model. While coarse-grained models of cellulose crystallites abound,^{39–42} they remain too complex for the simulation of large numbers of MFs over micro-mechanical timescales. For this reason, we exclude all atomistic detail and estimate the bonded parameters from experimental measurements of MF and BC fibre stiffness, assuming an aspect ratio, $\lambda = L/D = 100$, where L is fibre length and D is the bead diameter (Fig. 1). The total energy of the system is written as

$$e = e_T + e_B + e_{vdw}, \quad (1)$$

where $e_T + e_B$ and e_{vdw} are tensile, bending, and non-bonded interaction parameters, respectively. The tensile energy is characterised by

$$e_T = \frac{1}{2} k_T (r - r_0)^2. \quad (2)$$

k_T is a tensile spring constant, r is the distance between the centres of adjacent beads and r_0 the equilibrium distance, set here as equal to D (Fig. 1). Three-point nano-indentation of untreated *Acetobacter xylinum* BC ribbons produced $E_c = 78 \pm 17$ GPa,⁴³ close to the value reported by Tanpichai *et al.*²⁹ using Raman spectroscopy (RS) measurements (79–88 GPa). In contrast, Hsieh *et al.*⁴⁴ found a considerably higher modulus for these ribbons (114 GPa) from RS experiments, closer to that

Extraction of tension and bending energy from macroscopic parameters

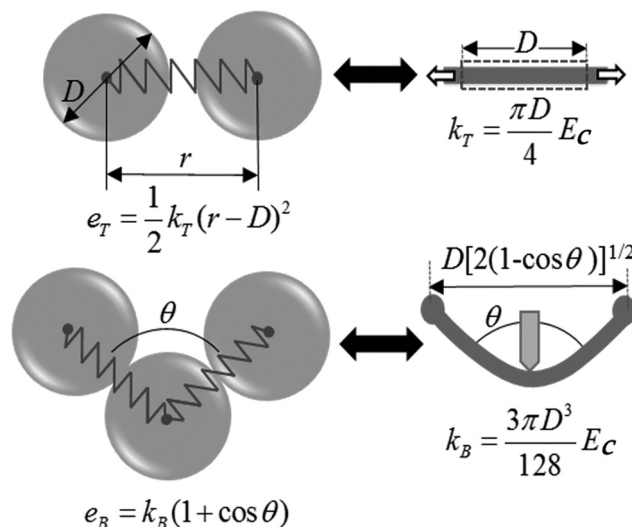


Fig. 1 Cellulose MFs are modelled as semi-flexible chains using a bead-spring approximation. Parameters characterising tension (e_T) and bending energy (e_B) are obtained by comparing the bead-spring system with a continuous beam under extension (top) and three-point bending (bottom), respectively, assuming a circular cross section area. Derivation of the equations for k_T and k_B are described in the Appendix.

independently estimated by Iwamoto *et al.*⁴⁵ and Šturmáková *et al.*⁴⁶ for a single tunicate MF from atomic force microscopy and RS, respectively (~ 145 GPa). This led Hsieh *et al.*⁴⁴ to conclude that their measurements were mostly reflecting the crystalline domains of BC ribbons. Indeed, neutron scattering measurements have shown *Acetobacter xylinum* BC ribbons to contain significant domains of both paracrystalline and amorphous cellulose,²⁷ in comparison with the highly crystalline plant MFs. Thus, we will assume that $E_c = 78$ GPa for BC ribbons and $E_c = 145$ GPa for a typical MF for our quantitative analysis.

In the linear elastic region, k_T in eqn (2) can be related to E_c through (see Appendix).

$$k_T = \frac{\pi D}{4} E_c. \quad (3)$$

The bending energy is characterised by a cosine potential,

$$e_B = k_B (1 + \cos \theta), \quad (4)$$

where θ is the angle between three consecutive beads and k_B the bending energy. Equating eqn (4) to the energy required to produce an equivalent beam bending gives (see Appendix):

$$k_B = \frac{3\pi D^3}{128} E_c, \quad (5)$$

Elementary MFs are thought to be only 2–3 nm in diameter,^{47,48} but they are often found forming thicker aggregates (up to ~ 10 nm) in primary cell walls.⁴⁹ Here, we will use $D = 2.74$ nm (for a 18 cellulose units MF) for our quantitative

estimates, and allow thicker fibrils to form spontaneously. Similarly, *Acetobacter xylinum* BC ribbons have been reported to have thicknesses that vary widely, between 40 and 100 nm.^{29,43,44} Thus, we will take $D = 30$ nm as the elemental mechanical unit in BC-based CHCs.

The non-bonded interactions between beads were modelled through the Lennard-Jones (LJ) potential,

$$e_{\text{vdW}} = 4\varepsilon \left[\left(\frac{D}{r} \right)^{12} - \left(\frac{D}{r} \right)^6 \right], \quad (6)$$

where ε is the LJ energy parameter. Let us first consider the value of ε between cellulose MFs. Using MD simulations, Oehme *et al.*⁵⁰ determined that the potential of mean force associated to the desorption of two cellulose MFs is a complex function of MF–MF separation and depends on the faces of cellulose crystallites directly interacting. They estimated desorption energies ranging between 14.7 kcal mol^{−1} and 61.2 kcal mol^{−1} for two parallel MFs with a DP = 20 (DP = degree of polymerization). If the bead diameter D in our CG model of a MF is 2.74 nm in diameter (which corresponds to ~ 5 DP), then Oehme *et al.*⁵⁰ model is ~ 4 beads in length. Thus, the interaction energy between beads should be taken as $\varepsilon \sim 3.67$ kcal mol^{−1} and 15.3 kcal mol^{−1}. For BC cellulose ribbons, we consider that the interaction strength between beads is proportional to the bead cross-section area. That is, $\varepsilon_{\text{ribbon}} = f\varepsilon_{\text{MF}}$, where $f = (D_{\text{ribbon}}/D_{\text{MF}})^2$ and the subscript indicates the type of fibre.

Table 1 summarizes the typical parameter values. Unless otherwise indicated, we have set $k_{\text{B}}/\varepsilon = 300$ and $k_{\text{T}}/\varepsilon = 10^4$ throughout most of our simulations. We shall see that reducing $k_{\text{B}}/\varepsilon = 30$ (corresponding to a MF in Table 1) has a relatively minor impact on extensibility but reduces the peak stress in uniaxial tension experiments. We must stress, however, that the purpose of this model is entirely qualitative, and quantitative agreement can only be reasonably expected within an order of magnitude approximation.

No sliding friction is incorporated in the model; modelling and experiment have shown frictional effects to be fundamental in the micro-mechanics of paper.⁵¹ However, it is considerably more challenging to assess the role of friction in highly hydrated environments such as CHCs synthesised by either bacteria (pellicle) or plants (wall). For instance, MD simulations predict spontaneous twisting of MFs,^{42,52,53} inducing surface roughness to the MFs. On the other hand, XG has been

suggested to coat MF surfaces and thus create a thin film barrier lubricating cellulose–cellulose contacts.^{49,54} However, ¹³C solid-state NMR results raise doubts as to whether much of the cellulose surface is coated with XG.^{55,56} Given the ambiguity, only van der Waals (vdW)-driven contacts are explored in this work.

2.2 Simulation setup

To set up the simulation, the chain centres were randomly distributed within the simulation box and a random orientation initially selected. However, to force the fibres (MFs or ribbons) to preferentially orient in the “in-plane” direction (xy -plane), the probability of accepting a given orientation from the random distribution was set as

$$P(\phi_{xy}) = \min \left\{ \exp \left[\frac{1}{a} \left(\frac{\pi}{2n_{xy}} - \phi_{xy} \right) \right], 1 \right\}, \quad (7)$$

where ϕ_{xy} is the angle between the proposed orientation and the xy -plane, a is constant (set as 0.2) and $n_{xy} \geq 1$ is the 2D anisotropy: $n_{xy} = 1$ produces isotropic out-of-plane orientations and $n_{xy} = \infty$ yields a planar system. To promote alignment in the x direction, a second independent probability distribution is defined as

$$P(\phi_x) = \min \left\{ \exp \left[\frac{1}{a} \left(\frac{\pi}{2n_x} - \phi_x \right) \right], 1 \right\}, \quad (8)$$

where ϕ_x is the angle between the proposed orientation and the x axis, and $n_x \geq 1$ is the 1D anisotropy: $n_x = 1$ corresponds to in-plane isotropy and $n_x = \infty$ to MFs that are in almost complete alignment with the x axis. A schematic representation of how $n_x = 1$ or 3, and $n_{xy} = 1$ or 3 constrain MF orientations is presented in Fig. 2.

The mass density of cellulose ρ_c is related to the number density of MFs ρ_N , through and the cellulose volume fraction, v_c , by

$$\rho_c = \frac{\pi}{6} \rho_s \rho_N \lambda D^3 = v_c \rho_s, \quad (9)$$

where $\rho_s = 1.5$ g cm^{−3} is the skeletal density of cellulose.

Systems with 250 chains, each with 100 beads and various degrees of “out-of-plane” and “in-plane” anisotropy were allowed to relax in the NVT ensemble at $T^* = kT/\varepsilon = 1$ for 10^7 time-steps using a Langevin thermostat. Here, k is the Boltzmann constant, and T is absolute temperature. The resulting system was quenched to $T^* = 0.01$ in 10^3 steps and further

Table 1 Typical order of magnitude for the model parameters. Unless otherwise indicated, all simulations were performed assuming $k_{\text{B}}/\varepsilon = 300$. The last row corresponds to the values employed in a typical simulation run ($10^6 E_c D^3/\varepsilon$ is a derived quantity and does not need be set up)

Dimensional parameters	E (GPa)	D (nm)	L (nm)	$10^4 k_{\text{T}}$ (kcal mol ^{−1} nm ^{−1})	k_{B} (kcal mol ^{−1})	ε (kcal mol ^{−1})
Cellulose MF	145	2.74	274	4.5	219	3.67–15.3
BC ribbon	78	40	1000	8.8	5743	783–3260
Dimensionless parameters	$10^6 E_c D^3/\varepsilon$	D	L/D	$\langle 10^3 D k_{\text{T}}/\varepsilon \rangle$	$\langle k_{\text{B}}/\varepsilon \rangle$	ε
MF/ribbons	0.14	1	100	5.05	~ 30	1
BC ribbon	9.45	1	100	17.2	~ 300	1
Typical run	—	1	100	10	300	1

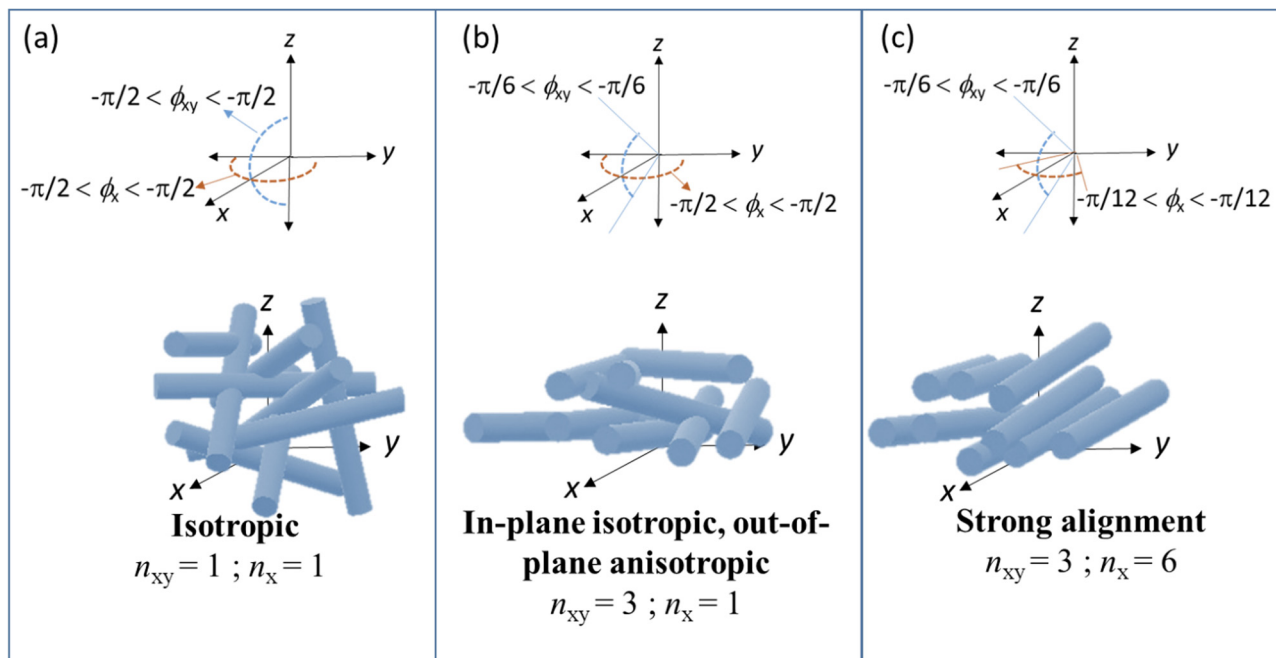


Fig. 2 Illustration of MFs alignment parameters n_{xy} and n_x . The angles ϕ_{xy} and ϕ_x indicate the most likely orientation of a beam with respect to the xy plane and the x axis, respectively. (a) $n_{xy} = n_x = 1$: isotropic distribution, (b) $n_{xy} = 3$ and $n_x = 1$: the beams are constrained with respect to the xy plane, with $-\pi/6 < \phi_{xy} < \pi/6$. There is no alignment with respect to the x axis. (c) $n_{xy} = 3$ and $n_x = 6$: strong alignment. The beams distributions are constrained with respect to the xy plane, between $-\pi/6 < \phi_{xy} < \pi/6$, and with respect to the x axis, with $-\pi/12 < \phi_x < \pi/12$.

equilibrated for 10^5 steps to render the system approximately athermal.⁴⁴ The timestep was set to $10^{-3}\tau$, where $\tau = (\epsilon/m)^{1/2}t/D$, t is time and m is bead mass. The systems were stretched at a strain rate $\dot{\gamma}_x = 10^{-3}\tau^{-1}$. The strain rate was set at a value low enough to guarantee that extension can be considered quasi-static. At every configuration, three system realisations were simulated, and the resulting stresses averaged.

Importantly, we do not assume a distribution of contact energies. In each simulation, the contact energy (characterized by k_B/ϵ) is fixed. The contact length between fibres is variable and left to the self-assembling process occurring spontaneously during the simulation. However, we can indirectly control the contact length by tuning the degree of alignment through parameters n_{xy} and n_x defined in eqn (7) and (8).

3. Results and discussion

3.1 System generation

Fig. 3b shows that the isotropic structure of semi-flexible chains predicted by the model, which compares favourably with experimental observations of the BC hydrogel network structure that is included in Fig. 3a.⁵⁷ Here, the resulting network for $n_{xy} = 1$ and $n_x = 1$ at a cellulose volume fraction, v_c , of 0.045 (*i.e.*, $\rho_c = 0.07 \text{ g cm}^{-3}$, considering the BC Ribbon parameters in Table 1). $n_{xy} = n_x = 1$ means the network is isotropic in both the “in-plane” (xy plane) and “out-of-plane” directions. The orientation distribution of the initial configuration is nearly identical to that of the quenched structure (inset), the latter measured over fragments of 10 consecutive beads. This indicates that model fibres

tend to only interact with their immediate neighbours and that parameters n_x and n_{xy} constitute a good measure of anisotropy for the final structures. Fig. 3c depicts a network with $\rho_c = 0.04 \text{ g cm}^{-3}$ and $n_x = 3$. The structural alignment is clear.

The concept of contact is not well defined, as some of the model fibres aggregate over their entire length. To establish a reasonable quantitative definition, bifurcation points were defined as points of contact initiation (Fig. 3d). This approach allows correlating the number of contacts n_c , anisotropy (n_x, n_{xy}) and stiffness. In the following section, we investigate the result of applying a uniform uniaxial strain rate on networks with different degrees of anisotropy, contact energy and interaction range.

3.2 The effect of anisotropy

Fig. 4 and 5 show the predicted response of the *in silico* networks to uniaxial extension at constant cross-sectional area and constant volume, respectively. The predicted dimensionless stress ($P^* = PD^3/\epsilon$) is plotted as a function of tensile strain for networks with two-dimensional anisotropy $n_{xy} = 3$ (eqn (7)), one-dimensional anisotropy ($n_x = 1, 3$ and 6 , eqn (8)) and density ($\rho_c = 0.05 \text{ g cm}^{-3}$, $\rho_c = 0.07 \text{ g cm}^{-3}$ and $\rho_c = 0.10 \text{ g cm}^{-3}$). The direction of extension coincides with the direction of alignment x . At any particular n_{xy} , increasing network density increases fracture stress, σ_{fracture} , and slightly decreases fracture strain, γ_{fracture} . On the other hand, there is a decrease in peak force and a substantial increase in γ_{fracture} as n_x increases at constant density.

Uniaxial stretching of highly concentrated polymers is typically carried out by controlling stress in the stretching directions, while setting the pressure to zero in the perpendicular

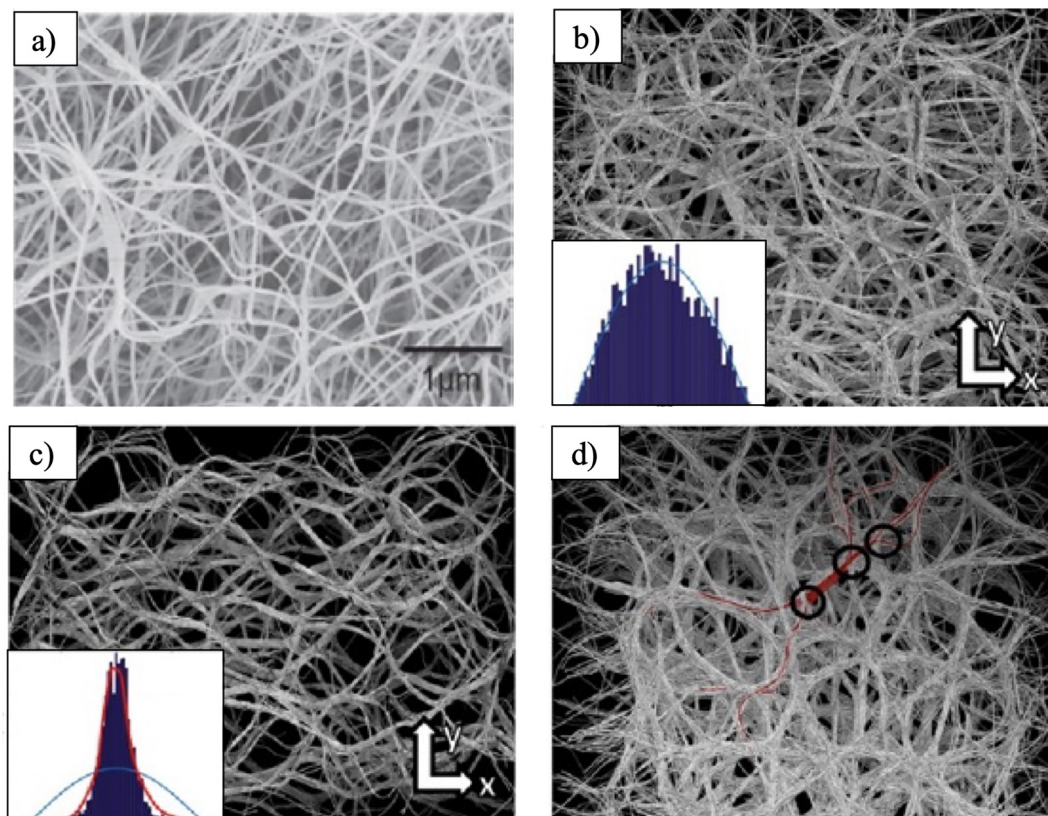


Fig. 3 Comparison of *in silico* cellulose networks with a BC hydrogel. (a) Micrograph of a BC hydrogel at a cellulose density $\rho_c = 0.07 \text{ g cm}^{-3}$. Reproduced with permission from ACS from Lopez-Sanchez *et al.*⁵⁷ (b) *In silico* cellulose network at a volume fraction $v_c = 0.045$ (i.e., $\rho_c = 0.07 \text{ g cm}^{-3}$, considering the BC ribbon parameters in Table 1), quenched from an initially isotropic distribution of extended MFs ($n_{xy} = n_y = 1$). The resulting xy orientation distribution is shown in the inset (bars) along with the ideal isotropic distribution (line). (c) *In silico* cellulose network at a density $\rho_c = 0.04 \text{ g cm}^{-3}$, with $n_x = 3$; the red line represents the initial ideal orientation distribution, the bars the orientation distribution of the quenched network, and the blue a perfectly random orientation distribution. (d) Definition of contact: the number of bifurcations emanating from a ribbon represent a contact initiation point. The ribbon in red has 6 contacts (there are 6 branches emanating from the encircled bifurcation areas).

direction.⁵⁸ This way, the Poisson ratio, ν , of the system will arise naturally from the simulation. In our simulations, the equilibrated system is under negative pressure. That is, at a pressure $P = 0$, the network would collapse into a polymer melt due to attractive vdWs interactions. Water is only implicitly present in the Langevin thermostat as a random (Brownian) and viscous force but exerts no outward pressure. In plant walls and BC hydrogels the negative pressure is counteracted by interstitial water and/or the interpenetrating non-cellulosic polysaccharides, avoiding structural collapse. Since we do not explicitly consider either water or non-cellulosic polysaccharides in this computational model, the network collapse is prevented by performing the simulations at:

(i) constant volume (Poisson's ratio $\nu = 0.5$), whereby it is assumed the network is incompressible, and thus the volume of interstitial "water" remains constant over the course of the stretching test (Fig. 4); or,

(ii) constant cross-section area, whereby it is assumed that, as the network is stretched, surrounding fluid is captured at a rate equal to the change in the volume of the network (Fig. 5). Therefore, the cellulose density constantly decreases.

It is reasonably expected that in the actual system, the moduli, fracture stress and fracture strain lie in between these two extremes. The second scenario, however, agrees with experimental observations on the compression of BC hydrogels: the cross-section area perpendicular to the compression direction remains constant and all changes in volume are a consequence of water flow normal to the compression direction.⁵⁷

Regardless of whether either constant volume or area is assumed in the simulation (Fig. 4 and 5), increasing alignment at constant density decreases peak force and increases extensibility. Since sliding of model fibres that are in contact is possible at a relatively low energy cost, the fibres are insignificantly stretched. That is, the contribution from the tensile strength of the fibres in our simulations is small at the examined strains, because the comparatively small values of ϵ/k_T and ϵ/k_B (the ratio between vdW and bonded interactions strength) make detachment a more favourable process. Greater alignment increases the contact length but reduces the number of contacts between misaligned MFs.

Fig. 6 shows previously reported experimental uniaxial stress-strain curves for (a) BC, BC-glucomannan (GM) and

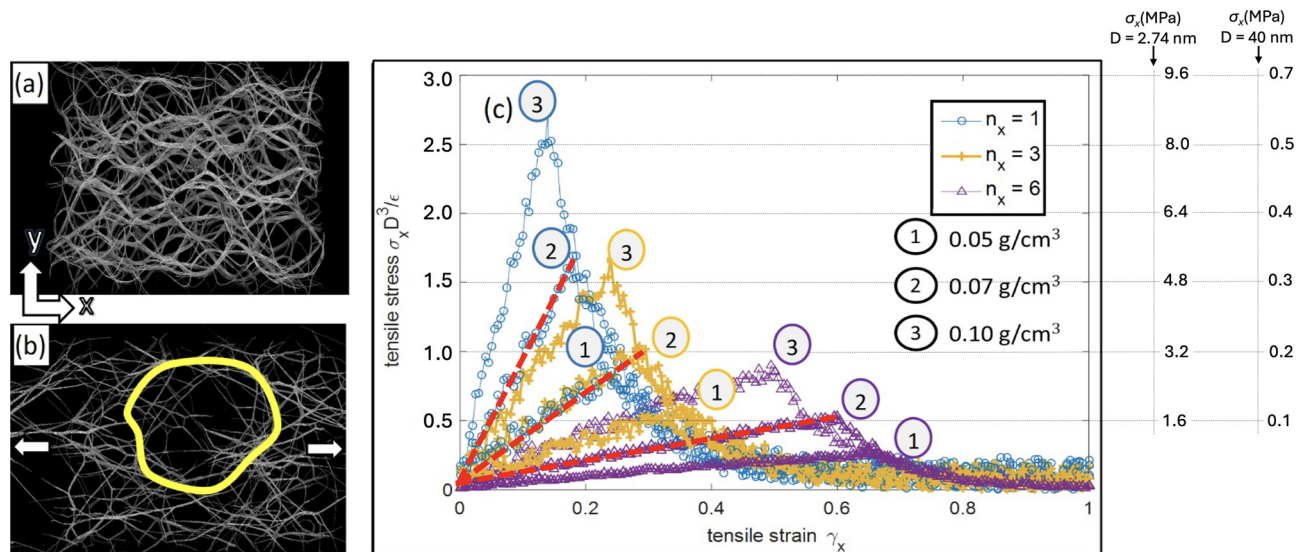


Fig. 4 Predicted mechanical response of the *in silico* networks to uniaxial extension at constant cross-sectional area. (a) *xy*-plane snapshot of an *in silico* cellulose network at a volume fraction $v_c = 0.045$ (i.e., $\rho_c = 0.07 \text{ g cm}^{-3}$, considering the BC ribbon parameters in Table 1), with $n_{xy} = 3$ and $n_x = 3$. Upon extension along the *x* axis at uniform cross-section area, an initial fracture zone appears at a tensile strain, $\gamma_x = 0.28$. (b) The ruptured structure at $\gamma_x = 0.38$; the initial fracture zone is encircled by the yellow curved line. (c) Dimensionless stress–strain for $\rho_c = 0.05, 0.07, 0.10 \text{ g cm}^{-3}$ for $n_{xy} = 3$ and $n_x = 1, 3$ and 6 . At each n_x , increasing peak forces correspond to increasing density. The red-dashed lines highlight the curves for $\rho_c = 0.07 \text{ g cm}^{-3}$ at the different alignment levels, which are further discussed in the manuscript. The scales on the right side represent the axial stress, σ_x , considering the parameters in Table 1 for a MF network ($D = 2.74 \text{ nm}$) and a BC composite ($D = 40 \text{ nm}$).

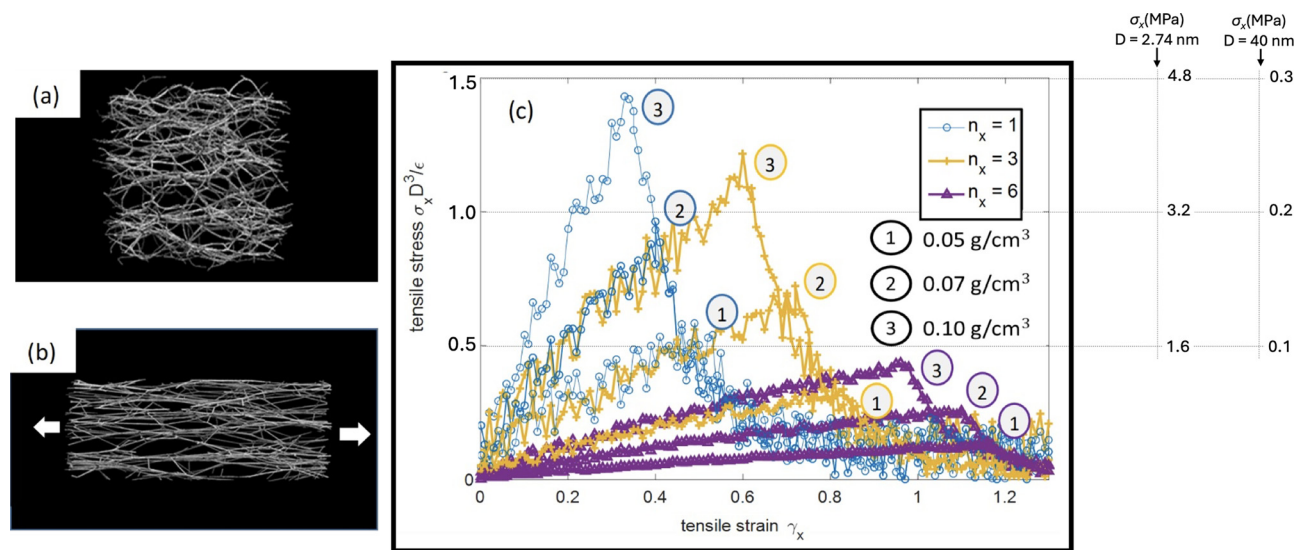


Fig. 5 Predicted mechanical response of the *in silico* networks to uniaxial extension at constant volume (a) *xy*-plane snapshot of a cellulose network having $\rho_c = 0.07 \text{ g cm}^{-3}$, $n_{xy} = 3$ and $n_x = 3$. (b) Uniaxial extension along the *x* axis at uniform volume, an initial fracture zone appears at a tensile strain $\gamma_x = 0.60$. (c) Dimensionless stress–strain for $\rho_c = 0.05, 0.07$, and 0.10 g cm^{-3} for $n_{xy} = 3$ and $n_x = 1, 3$ and 6 . Increasing peak forces correspond to increasing density. The scales on the right side represent the axial stress, σ_x , considering the parameters in Table 1 for a MF network ($D = 2.74 \text{ nm}$) and a BC composite ($D = 40 \text{ nm}$).

BC-XG CHC hydrogels (80:20 BC:XG and 49:51 BC:GM ratios, ρ_c not provided)⁴ (b) BC-pectin/ Ca^{2+} hydrogels ($\rho_c \approx 0.05 \text{ g cm}^{-3}$ and $\sim 47:53$ BC:pectin ratio)⁵⁹ and (b) BC-XG hydrogels before and after enzymatic hydrolysis ($\sim 80:20$ BC:XG, ρ_c not provided).²⁶ Clearly, the reported impact of XG in both Zhang *et al.*⁴ and Chanliaud *et al.*^{26,59} is quite similar,

despite differences in the quantitative measurements (which may be due to differences in the specific hemicellulose concentrations and water content of the composites). In summary, it is observed that:

(i) adding GM, XG and GM in the composites increases γ_{fracture} (extensibility) while reducing the peak stress.

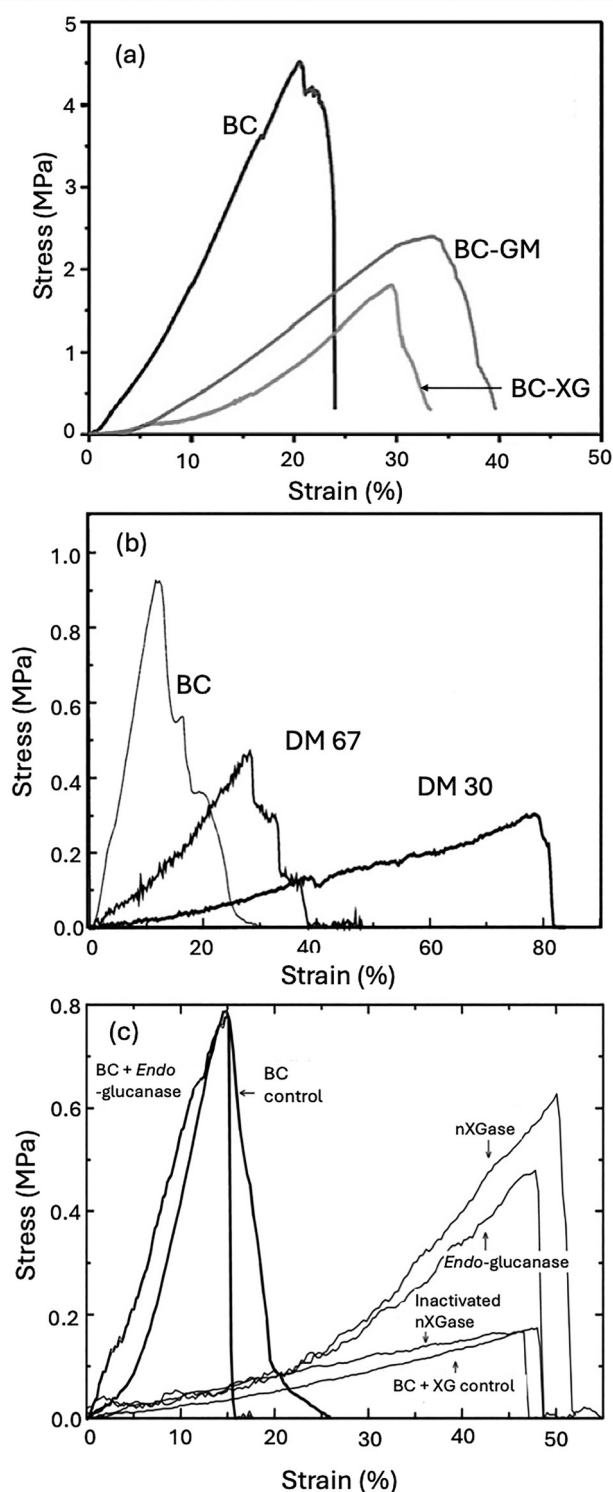


Fig. 6 Measured mechanical response of BC networks to extension. (a) Experimental stress–strain curves for (a) BC, BC–glucosaminan (GM) and BC–XG CHC hydrogels (80 : 20 BC : XG and 49 : 51 BC : GM ratios).⁴ Reproduced with permission of Wiley from Zhang *et al.*⁴ (b) BC and BC–pectin/ Ca^{2+} hydrogels (DM, degree of methyl esterification) with $\rho_c \approx 0.05 \text{ g cm}^{-3}$ and $\sim 47:53$ BC:pectin. Reproduced with permission of Wiley from Chanliaud *et al.*⁵⁹ (c) Experimental stress–strain curves for cellulose and cellulose–XG hydrogels before and after enzymatic hydrolysis of XG ($\sim 80:20$ BC : XG). Reproduced with permission of Wiley from Chanliaud *et al.*²⁶

(ii) The stress before fracture ($\sim 1 \text{ MPa}$) is significantly below that from stretching individual fibres ($\sim 1 \text{ GPa}$), indicating that cellulose fibres do not experience significant tension and that resistance to extension is concentrated on cellulose interactions (either direct or hemicellulose-mediated). And

(iii) treatment of XG–BC hydrogels with hydrolytic enzymes (XGase) shows that complete removal of XG–cellulose cross-links does not reduce extensibility but increases fracture strain (Fig. 6c).

Comparing with Fig. 4 and 5, it is apparent that pectin, XG and GM have a similar effect to that of increasing n_x anisotropy (fibre alignment). There are, however, both qualitative and quantitative differences between the model uniaxial tension curves and the experimental ones. First, the predicted stresses are lower than those in Fig. 6 (see right side of Fig. 4). However, the order of magnitude matches that in Fig. 6b and c, which is notable given that the model is parameterized with respect AFM to physical measurements as opposed to low-level atomistic simulations. Also, the stress decay after fracture is much more prolonged in the simulated networks. This is most likely an artifact from the simulations, as detached fibres may rapidly form new links with adjacent fibres. This likely does not occur during experimental runs.

Let us first consider the pure BC hydrogels in Fig. 6. These hydrogels were shown to be isotropic perpendicular to the extension direction.^{4,26,59} Thus, they are comparable to the $n_x = 1$ case in Fig. 4 and 5. The experimental γ_{fracture} varied between 15 and 20%, similar to that in Fig. 5 (constant cross-section area simulations), suggesting that a Poisson ratio $\nu \sim 0.0$ may be more representative of these systems, in accordance with the findings reported by Lopez-Sanchez *et al.*⁵⁷ Thus, we will focus on $\nu = 0.0$ results (Fig. 4) for the discussion below.

Let us focus on the curves for the $\rho_c = 0.07 \text{ g cm}^{-3}$ systems at $n_x = 1$, $n_x = 3$ and $n_x = 6$ (highlighted with the dashed lines in Fig. 4). The reduction (-34%) in fracture stress (σ_{fracture}) and increase in γ_{fracture} (0.17 to 0.28) when n_x increases from 1 to 3 is comparable to that in the CHC hydrogel containing DM 67 pectin (Fig. 6b), for which σ_{fracture} decreases by $\sim 50\%$ and γ_{fracture} increases from 0.15 to 0.27 with respect to the cellulose-only hydrogel. On the other hand, the changes depicted in Fig. 4 between $n_x = 1$ and $n_x = 6$ (-67% for σ_{fracture} and 0.17 to 0.60 for γ_{fracture}) are comparable to those arising from incorporating DM 30 pectin to BC in Fig. 6b (-75% for σ_{fracture} and 0.15 to 0.80 for γ_{fracture}), or from incorporating XG to BC in Fig. 6c (-75% for σ_{fracture} and 0.15 to 0.60 for γ_{fracture}).

Based on these observations, we hypothesise that enhanced reorientation during deformation may occur in the composite systems due to weaker “effective” contact forces between cellulose ribbons, enabling higher extensibility with respect to pure cellulose. In DM 30 composites, the structured pectin network creates pores and constraints, leading to a cellulose fibril arrangement with fewer interfibril contacts. For DM 67 pectin, the weaker pectin network allows cellulose fibrils to form a denser, more interconnected network, resulting in higher stiffness and lower extensibility compared to the DM 30 case. Nonetheless, both tend to produce lower σ_{fracture} and higher γ_{fracture} than pure cellulose hydrogels.

The experimental results in Fig. 6a for BC-GM and BC-XG CHCs support the enhanced reorientation hypothesis, despite clearly displaying higher fracture stresses than those in Fig. 6b and c (possibly due to higher hydrogel densities). We note that both XG and GM reduce fibre crystallinity (by 10 to 20%), impacting the tensile strength of the fibres and leading to lower tensile stresses. However, this is likely only a partial explanation, given that the fracture stress is many orders of magnitude below the modulus of individual fibres.⁴

Enhanced reorientation could result from weaker effective contact energy between individual fibres. Recent AFM nanotribological measurements consisting of pulling individual ribbons from microscopic BC and BC-XG hydrogels have shown that, statistically, the presence of XG weakens the adhesive fibre-fibre contact force. Indeed, using a cohesive zone model to interpret their results, it was concluded that the direct adhesive contact energy is (see Table 6.1 in Dolan⁶⁰ for $l = 0.3$ nm):

- ~ 60 mJ m⁻² between cellulose ribbons, which we define here as hard contacts.
- ~ 20 – 30 mJ m⁻² between cellulose ribbons in presence of XG, which we define here as soft contacts.

MF surface area is ill-defined in a bead-spring model but could roughly be assumed to correspond to the projection of the beads onto a plane ($\pi D^2/4$, $D = 2.74$ nm). In that case, the values above correspond to 51 kcal mol⁻¹ per bead for hard contacts, and 21 kcal mol⁻¹ per bead for soft contacts. We note, however, that these are average values and nanoscratching experiments produce a wide distribution of adhesion energies, as the interaction strength between a pair of cellulose fibres may vary widely depending on either the specific surfaces in contact or the adhesion area.

The behaviour observed in Fig. 6c upon treatment of XG-BC hydrogels with hydrolytic enzymes is consistent with a system in which the presence of XG-cellulose cross-links induces local alignment on the unconstrained structure: their removal replaces soft contacts by hard contacts without necessarily affecting the initial alignment, thereby preserving the original number of contacts. In planta, where cellulose is deposited under turgor-driven stress it is possible that pectin not only makes re-orientation easier but also promotes alignment of newly synthesised MFs.

Fig. 7 shows the predicted relative variation in (dimensionless) Young's modulus in the x (E_x) and y (E_y) directions with alignment (n_x) for $n_{xy} = 3$ at $\rho_c = 0.07$ g cm⁻³. The maximum Young's modulus is that of an in-plane isotropic network, $n_x = 1$, in which case $E_x = E_y$. For $n_x > 1$, the modulus in the axis of alignment, E_x , is always higher than that in the transverse axis, E_y , but both decrease as the axial anisotropy n_x increases.

These simulation results provide a complementary viewpoint on the phenomenon of strain hardening experimentally observed in both BC hydrogels and plant cell walls, which may be considered to be approximately two-dimensional CHCs aligned along one main axis: when a force is experienced in the z direction (e.g. by the wall of the plant cell during volumetric expansion), the external force induces in-plane reorientation, increases isotropy (i.e., reducing n_x) and consequently, the

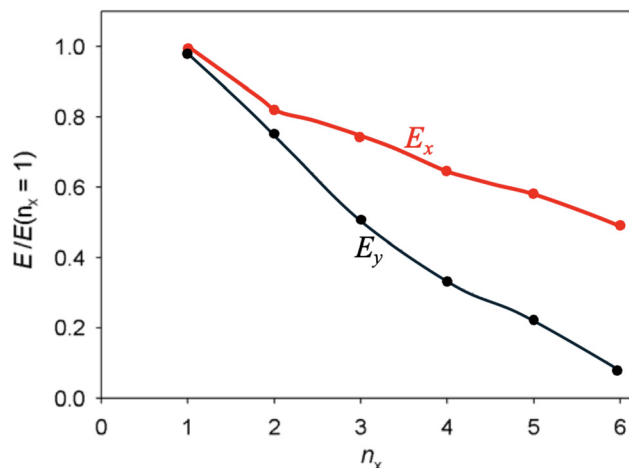


Fig. 7 Prediction for the variation in Young's modulus with n_x relative to an in-plane isotropic network ($n_x = 1$), for $n_{xy} = 3$ at $\rho_c = 0.07$ g cm⁻³. The modulus in the axis of alignment, E_x , is always higher than that in the transverse axis, E_y , but both decrease as n_x increases.

Young's modulus in both extension (x) and transverse (y) directions increases.

There are significant differences with regards to the tensile modulus of primary walls, which is strongly dependent on polymer composition, arrangement and hydration levels. For instance, Zamil *et al.*^{61,62} measured uniaxial stress in isolated onion cell wall fragments in “never dried” samples, carefully excluding the middle lamella. The mean E_x (major growth direction) and γ_{fracture} were ~ 3.7 GPa and 3.0%. Nezhad *et al.*¹⁹ measured longitudinal and transverse moduli in *Camellia japonica* pollen tubes *via* lab-on-a-chip bending tests and finite element analysis, reporting E_x/E_y values between 0.3 and 0.4 GPa. Pereira *et al.*⁶³ employed a microplate-based bending technique to measure the Young's modulus of *in vivo* root hair cell walls in *A. thaliana*. Wild-type root hair walls exhibited an average Young's modulus of 0.4 ± 46 GPa (assuming a cell wall thickness of 250 nm), while erulus mutants, with thicker walls (500 nm), showed a significantly lower modulus of 0.073 ± 21 GPa. On the other hand, recent *in vivo* AFM measurements by Tsugawa *et al.*⁶⁴ were able to separate turgor contributions from wall elasticity, a typical issue *in vivo* measurements. They report average in-plane moduli values in onion epidermal cells of approximately 0.4 GPa.

The cellulose density in primary walls can vary significantly between species and tissues. Considering that (i) hydration levels are typically around 80%, (ii) usually cellulose constitutes $\sim 20\%$ of the dry mass (e.g. in onion epidermis cell walls) and that (iii) both cellulosic and non-cellulosic polysaccharides have a skeletal density of ~ 1.5 g cm⁻³ (thus, the density of the wall can be assumed to be ~ 1.3 – 1.4 g cm⁻³),^{65,66} we conclude that $\rho_c \sim 0.05$ g cm⁻³– 0.07 g cm⁻³. Consequently, for an in-plane isotropic wall and assuming $\nu = 0$, the parameters in Table 1 predict a modulus of ~ 65 MPa. Consequently, the current model tends to underpredict the in-plane modulus in primary walls by a factor of ~ 6 . On the contrary, the model produces a modulus comparable to that in hydrated

BC hydrogels (~ 5 MPa) when assuming $n_x = 1$,⁵⁹ and BC/XG composites assuming $n_x = 6$ (~ 0.4 MPa).^{25,26,59} This may be indicative that BC-based analogues tend to understate the mechanical role of fibre stretching, as inter-fibre adhesion may be easier to overcome than in actual walls (*i.e.*, contacts tend to be stronger in planta³⁷).

3.3 Effect of contact energy

The predicted effect of contact energy (ϵ/k_B term) between model fibres in the network during extensional deformation is shown in Fig. 8. Here, we depict the stress *versus* strain curves at several values of ϵ/k_B for a low-density network ($\rho_c = 0.04$ g cm⁻³), $n_{xy} = 1$ and $n_x = 1, 3$ and 6 with uniform cross section area ($\nu = 0$). The wide range of values of ϵ/k_B explored in this Fig. 8 (0.3, 0.03 and 0.003) include contact energies below and above those established for the average soft (20–30 mJ m⁻², $\epsilon/k_B \sim 0.01$ for BC fibres) and hard contacts (60 mJ m⁻², corresponding to $\epsilon/k_B \sim 0.10$ for BC fibres), offering a complete overview of the expected micromechanical behaviour of CHCs under uniaxial tension. Thus, $\epsilon/k_B \sim 0.3$ would correspond to a system in which most contacts are hard (*i.e.*, fibres interact through direct hydrogen bonding), $\epsilon/k_B \sim 0.003$ to a system in which contacts are soft (*i.e.*, all fibre interactions are implicitly mediated by hemicelluloses) and $\epsilon/k_B \sim 0.03$ to a system in between these two extremes (a fraction of both types of contacts is present).

Reducing ϵ/k_B over several orders of magnitude ($\epsilon/k_B = 0.03, 0.03$ and 0.003) is shown to reduce σ_{fracture} substantially, but does not significantly increase extensibility. This agrees with computational studies of crack formation in epoxy glasses with no cross-linking,⁶⁷ which concluded that fracture strain at a given density was related to the distance required to overcome the LJ particle interaction potential between nearest neighbours. The predicted fracture strain is strongly influenced by the degree of anisotropy for all values of ϵ/k_B . This occurs because the degree of network alignment determines how long

the fibres must slide past each other to overcome the LJ attractive interactions between them.

3.4 The contacts hypothesis

The predicted influence of structural network variables on the stiffness of the model cellulose fibre network are shown in Fig. 9. Dimensionless Young's modulus is presented as a function of the average number of contacts per ribbon (n_c) in networks with different densities and degrees of anisotropy, at a uniform contact energy $\epsilon/k_B = 300$. The figure suggests that, ultimately, n_c is the structural parameter responsible for changes in Young's modulus. Variations in density and/or anisotropy mostly led to variations in the modulus by altering the value of n_c .

Decreasing the contact energy reduces the predicted value of E_x , as noted from the stress–strain curves at several contact energies in Fig. 8, but E_x still increases with n_c . Thus, E_x can predominantly be considered a function of two parameters:

- one is structural (n_c) that lumps density and orientation effects; and
- the second is related to the intrinsic interaction potential, the contact energy ϵ/k_B and, potentially, the interaction range.

These observations provide insights into the how cellulose networks could respond to mechanical forces in nature. We suggest that plants could modify n_c to compensate for an inadequate ratio of “soft-to-hard” contacts by, for example, either increasing MF alignment or locally modifying hemicellulose concentrations. Similarly, this ratio could be modified by design when synthesising bacterial CHCs to fine-tune their micromechanical properties, which may be of interest in the development of biomedical scaffolds.

4. Conclusions

This study examines the hypothesis that the mechanical response of cellulose networks in nature can be reasonably captured using a coarse-grained molecular dynamic model with

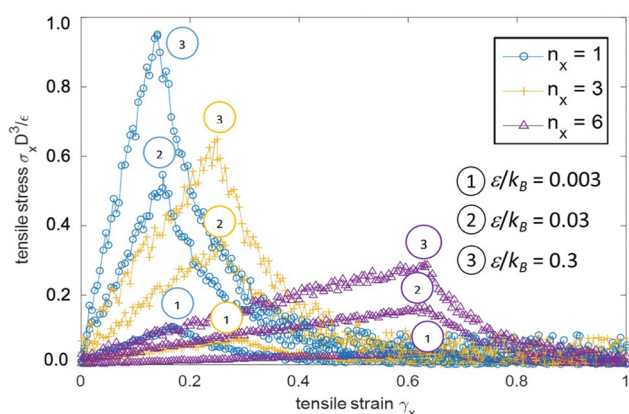


Fig. 8 Predicted mechanical response to uniaxial extension for model cellulose networks as a function of contact energy and anisotropy. Stress–strain curves at $\epsilon/k_B = 0.3, 3 \times 10^{-2}$ and 3×10^{-3} for $\rho_c = 0.05$ g cm⁻³, $n_{xy} = 1$ and $n_x = 1, 3$ and 6 (at constant cross-section area). Clearly, reducing ϵ/k_B reduces the peak stress but does not significantly increase extensibility.

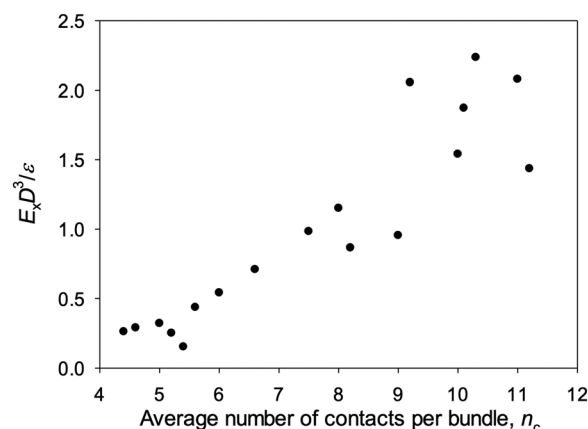


Fig. 9 Prediction of the dimensionless Young's modulus as a function of average number of contacts per ribbon (n_c) in networks with different densities (0.07 g cm⁻³ < ρ_c < 0.21 g cm⁻³), and degrees of in-plane and out-of-plane anisotropy ($1 < n_x, n_{xy} < 6$) at a contact energy $\epsilon/k_B = 300$.

bead-spring chains used to model cellulose fibre networks. It is found that the model qualitatively reproduces the mechanical behaviour of BC hydrogels under uniaxial tension.

At a prescribed network density, the fully stretched fibres are placed within the simulation box using a Monte Carlo-based algorithm to generate a desired level of anisotropy in selected directions. Fibre interact through van der Waals forces and, upon equilibration at high temperature, the network is quenched, and entropic mechanical effects minimised. It was found that the contact forces between model fibres could be ‘softened’ by reducing the contact energy and increasing the contact length of fibre–fibre interactions. As the contact forces between fibres are softened, the modulus of the network decreased but the network was able to extend to increasingly greater values of strain. In particular, soft contacts facilitate fibre rotation about contact points (re-orientation), promoting extensibility. When all contacts are energetically equivalent, the predicted Young’s modulus is found to be a function of the average number of contacts per ribbon (n_c). The influence of other structural properties such as the degree of anisotropy and network density on the Young’s modulus arises primarily from their impact on modifying n_c .

The softening of contact forces was qualitatively predicted to correspond to the effect of polysaccharides on the measured mechanical response of bacterial cellulose networks. That is, a loss of stiffness and increased extensibility under tensile forces is observed for pectin–BC and XG–BC hydrogel composites in comparison with their purely cellulosic counterparts. This is hypothesised to correspond to the model because a decrease in adhesion between fibres by XG has also been measured using AFM.

The current model may also provide insights on our understanding of the mechanics of plant walls. We suggest that polysaccharide mediated cellulose–cellulose interactions (most significantly XG in type 1 walls) by decreasing the contact energy between fibres, and we suggest this may be the mode of action rather than the formation of tethers previously speculated in the literature. Thus, we suggest direct cellulose–cellulose contacts could be classed as a “hard contact”, while those mediated by polysaccharides could be classed as a “soft contact”.

This work represents an initial exploration into modelling the experimental behaviour of cellulose hydrogel composite systems at the mesoscopic level through molecular simulation methods. While our model qualitatively captures essential features of fibre–fibre interactions and the resulting network mechanics, it purposely omits other structural components—such as non-cellulosic polysaccharides and dynamic bonding–debonding processes—which will be critical in future models. These extensions promise to broaden our understanding and yield more faithful representations of the complex architectures found both in planta and cell wall CHC hydrogel analogues, paving the way for increasingly predictive and robust simulations.

Future studies should also focus on the effect of varying the ratio of “soft to hard” contacts in the model. We hypothesise

that when this ratio is above its optimal value (such as in XG-depleted mutants),¹² plants can decrease n_c by purposefully increasing fibril alignment and hence, providing the lost flexibility through purely architectural means. Finally, it is assumed here that fibres experience friction-free sliding. The presence of friction and the lubrication effect that polymer coating could provide has not been investigated here but could readily be implemented within the current framework.

Conflicts of interest

There are no conflicts of interest to declare.

Data availability

Input data for this article, including the script to generate the system configurations, lammmps input file and sample configuration file are available at <https://doi.org/10.6084/m9.figshare.28652705>.

Appendix

Tension coefficient, k_T

To derive the parameter k_T for the bond interaction between adjacent beads, we assume that individual fibres experience only small strains during the simulations, and that the elastic stress is exclusively due to the stretching of bonds between beads. In that case, we have from eqn (2) that

$$\frac{de_T}{dr} = k_T(r - r_0) = \frac{\pi D^2}{4} E_c \varepsilon, \quad (10)$$

where $\varepsilon = (r - r_0)/D$ (see Fig. 1, top) and E_c the tensile modulus of the fibre. Solving for k_T yields

$$k_T = \frac{\pi D}{4} E_c \quad (11)$$

Bending coefficient, k_T

Consider Fig. 1, bottom. We assume that angular bending of a three beads system can be equated to the three-point bending of a thin filament of length $2D$. The bending force F for a filament bent through its center varies with the indentation depth as

$$F = \frac{48 E_c I}{L^3} \delta, \quad (12)$$

where I is the moment of inertia. For a beam of circular cross section

$$I = \frac{\pi D^4}{64}, \quad (13)$$

The bending energy $e_B(\delta)$ is obtained by integrating the force in eqn (12) ($= \int_0^\delta F(\delta') d\delta'$). Combining eqn (12) and

(13) (with $L = 2D$) and performing this integration yields

$$e_B = \frac{3\pi D^3}{128}(1 + \cos \theta) \quad (14)$$

equating eqn (4) and (14) produces

$$k_B = \frac{3\pi D^3}{128}E_c, \quad (15)$$

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